

LunarAtlas: From PDS4 Archives to Analysis-Ready Spectra — A Reproducible Infrastructure for Chandrayaan-3 LIBS Level-1 Data

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Abstract

We present **LunarAtlas**, a reproducible data-processing infrastructure that transforms publicly available Chandrayaan-3 Laser-Induced Breakdown Spectroscopy (LIBS) Level-1 (L1) products into cleaned, machine-accessible, long-form spectral records suitable for quantitative lunar science. Starting from calibrated L1 tables released through ISRO’s PDS4-compliant archive, LunarAtlas implements a transparent Python pipeline that: (i) parses XML metadata; (ii) reshapes wide-format Chandrayaan-3 L1 tables—in which 2049 wavelength channels appear as column headers spanning 164.35 nm to 878.26 nm—into normalised per-wavelength long-form records; (iii) identifies and correctly pairs plasma and background measurements using mission flag columns (**Laser_Fire_Status**, **Force_Reset_Status**); and (iv) performs physically motivated background subtraction, $I_{\text{clean}}(\lambda) = \max(0, I_{\text{plasma}}(\lambda) - I_{\text{background}}(\lambda))$, with non-physical negatives clamped to zero. Each processed file receives an MD5 checksum and an algorithm version record. Applied to `ch3_lib_002.20230825T104221.00.11.csv`, the pipeline yields four cleaned measurements (**Measurement IDs 1–4**), each with 2094 wavelength channels and 19 metadata columns. A baseline suppression factor of ~ 7.1 and a high-frequency variance increase of ~ 3.9 are observed, validating the physical correctness of the pipeline. The fraction of negative samples before clamping ranges from 0.5% (ID 3, high-quality plasma) to 76.8% (ID 2, near-null acquisition), demonstrating that the **Measurement ID** is essential for identifying shot-to-shot variability that would be hidden in bulk averaging. LunarAtlas further provides a PDS4-aware PostgreSQL schema, adaptive min–max downsampling with a mathematically defined zoom-saturation criterion, and a versioned API with sub-500 ms response times. By establishing a “single source of scientific truth,” the system offers a replicable methodology for Chandrayaan-4, Artemis-era VIPER, and Commercial Lunar Payload Services (CLPS) LIBS instruments.

Keywords: Laser-Induced Breakdown Spectroscopy; Chandrayaan-3; Lunar Regolith; PDS4; Data Cleaning; Spectral Reproducibility; Background Subtraction; NIST Validation; Measurement ID.

1. Introduction

1.1 The Promise of Lunar In-Situ Spectroscopy

The elemental composition of the lunar surface encodes four billion years of planetary evolution—from the primordial magma ocean, through impact gardening, to ongoing solar-wind implantation of hydrogen and helium. Accessing this record has long depended on remote sensing and the painstaking laboratory analysis of returned samples. Laser-Induced Breakdown Spectroscopy (LIBS) offers a fundamentally different path: a pulsed laser focused on a surface target ablates a few nanograms to micrograms of material, generating a transient plasma whose characteristic optical emission encodes the elemental abundances of the ablated material with no sample preparation and no contact requirement [1, 2].

The technique was first demonstrated in a planetary setting aboard NASA’s Mars Science Laboratory (MSL) rover *Curiosity* in 2012, where the ChemCam instrument acquired more than 930 000 single-shot spectra from 3 500 unique rock and regolith targets in Gale Crater [2, 3, 4]. The global community subsequently adopted LIBS for the SuperCam instrument on NASA’s

Perseverance rover and the MarSCoDe instrument on China’s Zhurong rover, cementing LIBS as the *de facto* in-situ geochemical sensor for planetary surfaces [5, 7].

Against this backdrop, the successful soft landing of India’s Chandrayaan-3 mission near the lunar south pole on 23 August 2023 marked a milestone of particular significance: the *first deployment of LIBS on the Moon* [9]. The Pragyan rover carried a miniaturised LIBS instrument alongside an Alpha Particle X-ray Spectrometer (APXS) [10]. Operating in the near-vacuum lunar environment introduces plasma dynamics fundamentally different from the ~ 7 mbar Martian atmosphere where ChemCam was optimised [13], underscoring the scientific importance of correctly handling and interpreting the resulting spectral data.

1.2 The Data Accessibility Gap

ISRO released the Chandrayaan-3 LIBS data as calibrated Level-1 (L1) products conforming to Planetary Data System version 4 (PDS4) standards [11]. PDS4 is a rigorous archival format: each data product consists of a self-describing XML label paired with a tabular data file, ensuring long-term preservation and provenance.

However, the Chandrayaan-3 L1 tables present a

structural challenge that is not immediately obvious from the PDS4 documentation: the intensity data are encoded in a *wide format* in which the 2049 wavelength channels (spanning 164.35 nm to 878.26 nm) appear as column *headers*, with their numeric wavelength values used directly as column names (e.g., 164.35, 164.74, ..., 878.26). Each row of the table corresponds to a single measurement acquisition, and the first six columns store mission-level metadata: **Time**, **Measurement Count**, **Operation Mode**, **Measurement Type**, **Force Reset Status**, and **Laser Fire Status**. The final eight columns record instrument housekeeping parameters: delay time, integration time, number of pulses, X-factor, laser energy, laser pump current, pulse repetition rate (PRR), and on/off status.

This wide-format structure is not directly amenable to:

1. relational database storage and SQL-level spectral queries;
2. pairing of plasma and background shots, which are interleaved as separate rows in the same table distinguished only by the **Force_Reset_Status** and **Laser_Fire_Status** flag columns;
3. consistent background subtraction across all wavelength channels in a single operation;
4. downstream peak detection, NIST line matching, or chemometric modelling.

Furthermore, without explicit file integrity checks and algorithm-version records, any reprocessing of these data is difficult to audit or reproduce across research groups.

1.3 Research Question and Contributions

The central question addressed by this paper is:

How can Chandrayaan-3 LIBS Level-1 data—as released through ISRO’s PDS4-compliant archive—be cleaned, normalised, and structured into a reproducible, queryable, and machine-accessible resource without compromising spectral fidelity or peak integrity?

This paper makes the following contributions, ordered by scientific importance:

- (C1) **Level-1 cleaning pipeline.** A transparent, step-by-step Python pipeline that ingests Chandrayaan-3 LIBS L1 tables, correctly identifies plasma versus background measurements using mission flag columns, reshapes wide-format data into normalised long-form spectra, and performs physically motivated background subtraction.
- (C2) **Measurement ID semantics.** A formal definition and implementation of the *Measurement ID*—the key identifier that links each cleaned spectral

record to its originating plasma shot, instrument configuration, and acquisition timestamp—with a full explanation of what it means, how it is derived, and why it is essential for reproducibility.

- (C3) **PDS4-aware data model.** A normalised PostgreSQL schema that stores cleaned spectral samples together with the complete mission hierarchy (mission → instrument → observation → measurement → wavelength → intensity), together with MD5 file checksums and algorithm-version metadata for every ingested file.
- (C4) **Zoom-aware visualisation.** An adaptive min-max downsampling formulation—mathematically defined in terms of zoom level and wavelength span—that preserves emission-line peaks and baseline structure across all zoom levels, with an analytic zoom-saturation criterion identifying when the system must fall back to raw-data mode.
- (C5) **Empirical cleaning validation.** Quantitative characterisation of the effect of background subtraction on a real Chandrayaan-3 L1 file, including baseline suppression factors and high-frequency variance behaviour, reported for the first time for this specific mission product.

1.4 Paper Organisation

Section 2 reviews related work. Section 3 describes the Chandrayaan-3 dataset and PDS4 context in detail. Section 4 presents the LunarAtlas architecture and data model. Section 5 details the Level-1 cleaning pipeline, including the critical Measurement ID concept. Section 6 covers the adaptive downsampling formulation. Section 7 describes the API and visualisation workflow. Section 8 presents cleaning results and validation. Section 9 discusses implications and limitations. Section 10 concludes.

2. Related Work

2.1 Planetary LIBS Instruments and Mission Heritage

The deployment of LIBS on planetary surfaces has a rich history beginning with ChemCam on MSL *Curiosity* [2, 3]. ChemCam’s three Czerny-Turner spectrometers cover 240–342 nm (UV), 382–469 nm (violet), and 474–907 nm (VNIR), and its Nd:KGW laser at 1067 nm has enabled remote analysis of targets up to 7 m distant. After more than 4 000 Martian sols, ChemCam has characterised chemostratigraphy in Gale Crater, detected boron and fluorine in-situ for the first time on Mars, and quantified hydrogen as a proxy for hydration [22, 21].

SuperCam on *Perseverance* extends ChemCam’s capability by integrating LIBS with Raman spectroscopy, time-resolved luminescence, and VIS-IR reflectance, as

well as an acoustic microphone for rock hardness assessment [5, 6]. China’s MarSCoDe instrument on Zhurong combines LIBS and short-wave infrared (SWIR) spectroscopy across 240–2400 nm and has characterised surface composition in Utopia Planitia [7, 8].

For lunar LIBS, earlier feasibility work demonstrated acceptable signal-to-noise in simulated vacuum conditions below 10^{-3} mbar [18, 19] and characterised hydrogen detection capabilities relevant to potential volatile deposits [20]. The Chandrayaan-3 LIBS instrument, a direct descendant of the design prepared for Chandrayaan-2, represents a miniaturised system weighing approximately 1.1 kg with power consumption below 1.2 W [10].

2.2 Planetary Data Systems and PDS4 Standards

PDS4 was adopted by NASA and partner space agencies to address limitations of the earlier PDS3 standard, particularly around self-describing data products and unambiguous semantic labelling [11]. In PDS4, every data product consists of an XML label—conforming to the PDS4 Information Model—paired with a data file in tabular, binary, or image format. The XML label explicitly specifies column names, data types, units, and fill values, making the format theoretically self-sufficient.

In practice, however, PDS4 archives are designed for long-term preservation rather than interactive analysis. They provide no relational abstractions, no query APIs, and no standard tooling for pairing observational products (e.g., matching plasma shots to their background acquisitions). Generic PDS4 readers such as the GDAL PDS4 driver treat the data as raster or tabular layers without encoding instrument-specific semantics [12]. Published analyses of Chandrayaan-3 data have typically relied on mission-specific scripts that are not versioned, archived, or shared with reproducibility in mind.

2.3 Spectral Data Processing and Background Subtraction

Background subtraction is a foundational step in LIBS data reduction. In Martian LIBS, ChemCam employs both passive background measurements (with the laser blocked) and active background estimation via wavelet-based continuum modelling [15]. A plasma shot typically involves 10–30 laser pulses, and the background is often acquired immediately before or after the plasma burst.

For ChemCam and SuperCam, extensive preprocessing pipelines have been developed, including Stationary Wavelet Transform (SWT) denoising with Daubechies-2 wavelets, Asymmetric Least Squares (ALS) baseline correction, and wavelength calibration against titanium emission lines [14, 15]. Machine learning approaches—partial least squares regression (PLSR),

convolutional neural networks (CNN), and transfer learning frameworks—have also been applied to ChemCam data for elemental quantification [15, 16, 13].

For Chandrayaan-3 specifically, the recent independent study by Sridhar et al. applied Savitzky-Golay polynomial filtering for denoising and Asymmetric Least Squares for baseline correction, and demonstrated identification of Mg II, Al I, Ca II, Ca I, Fe I, Si I, Ti II, Na I, and O I emission lines in the Chandrayaan-3 L0 data [17]. Yumoto et al. demonstrated detectability of hydrogen and felsic-to-basaltic compositional transitions in lunar-analog samples via pulsed LIBS, establishing line intensity ratios (Mg/Si, Ca/Al) as proxies for highland vs. mare regolith [20]. Knight et al. provided the fundamental physics baseline for vacuum LIBS, showing that laser-induced plasmas in pressures below 10^{-3} mbar produce realistically detectable emission from Si, Al, Mg, and Ca despite free plume expansion [18]. LunarAtlas complements this body of work by focusing on the L1 data tier, providing a formal data model, reproducibility infrastructure, and the Measurement ID concept that none of the above pipelines implement.

2.4 Downsampling for Interactive Spectral Visualisation

General-purpose time-series downsampling algorithms—including mean aggregation, median aggregation, Douglas-Peucker curve simplification, and the Largest Triangle Three Buckets (LTTB) method [23]—are widely deployed in web visualisation frameworks. These methods minimise visual distortion of the overall spectral envelope but do not provide any guarantee of retaining narrow emission lines, which may span only one to three wavelength channels. For LIBS spectroscopy, where a narrow Ca II doublet at 393–397 nm or an Mg II line at 280 nm can be the primary diagnostic signature of an element, this is scientifically unacceptable. The min-max strategy introduced in LunarAtlas specifically addresses this gap.

3. Chandrayaan-3 LIBS Dataset and PDS4 Context

3.1 Mission Overview

Chandrayaan-3 was launched on 14 July 2023 by the Indian Space Research Organisation (ISRO) and achieved a successful soft landing near the lunar south pole at 69.37°S, 32.35°E on 23 August 2023, at a site subsequently named *Shiv Shakti Point* [9]. This made India the fourth nation to soft-land on the Moon and the first to land in the high southern polar region, an area of substantial scientific interest owing to permanently shadowed regions that may harbour water ice and other volatiles [24, 25].

The lander, *Vikram*, deployed the *Pragyan* rover, which carried two in-situ science payloads: the LIBS instrument and the Alpha Particle X-ray Spectrometer

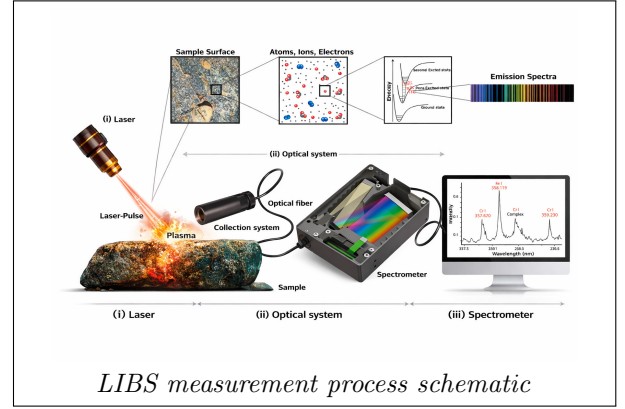


Figure 1: (a) The Pragyan rover on the lunar surface near Shiv Shakti Point, showing the LIBS instrument aperture (image credit: ISRO). (b) Schematic of the LIBS measurement principle: a focused laser pulse ablates material from the target surface, creating a transient plasma plume whose characteristic emission spectrum is collected by the onboard spectrometer. Background acquisitions (laser off) are interleaved with plasma acquisitions (laser on) to enable subtraction of ambient signal.

(APXS). The rover operated for approximately 14 Earth days—one lunar day—traversing a total distance of approximately 100 m before entering sleep mode ahead of the lunar night, from which it did not wake [13].

3.2 The Pragyan LIBS Instrument

The LIBS instrument aboard Pragyan is a miniaturised, low-mass design (approximately 1.1 kg, <1.2 W) developed at ISRO’s Space Physics Laboratory [10]. It employs a pulsed Nd:YAG laser focused on the lunar surface at close range and collects plasma emission with a spectrometer spanning the mid-ultraviolet to visible range. The system was designed to operate in the near-vacuum lunar environment, where the absence of atmospheric pressure means the LIBS plasma expands freely as an unconfined plume—a regime qualitatively different from Martian LIBS, in which the ~ 7 mbar CO_2 atmosphere confines the plasma and enhances emission intensity [18, 13].

The operational measurement sequence consists of alternating *background* acquisitions (laser not fired, recording ambient thermal and solar radiation) and *plasma* acquisitions (laser fired, recording plasma emission superimposed on background). This interleaved design allows the background to be subtracted from the plasma spectrum to isolate true LIBS emission. Fig. 1 illustrates the working principle and the Pragyan rover configuration.

The archival L1 data file analysed in this study is `ch3_lib_002_20230825T104221_00_11.csv`, acquired on 25 August 2023 at 11:35 UTC, approximately two Earth days after landing.

3.3 Level-1 Data Format: Wide-Table Structure

The Chandrayaan-3 LIBS L1 products are distributed in a wide-format tabular structure that departs significantly from the long-form schema typically expected

by relational databases or scientific Python workflows. Table 1 summarises the column organisation.

Table 1: Column organisation of Chandrayaan-3 LIBS Level-1 CSV files. The first six columns are metadata; the following 2049 columns are wavelength channels; the final eight columns are instrument housekeeping parameters.

Group	Contents
Metadata (6)	Time, Measurement Count, Operation Mode, Measurement Type, Force Reset Status, Laser Fire Status
Wavelengths (2049)	Column headers are wavelength values in nm (164.35 – 878.26); cells are intensity counts
Housekeeping (8)	Delay Time, Integration Time, Number of Pulses, X-Factor, Laser Energy, Pump Current, PRR, On/Off Status

Each row of the table represents a single measurement acquisition. The file `ch3_lib_002_20230825T104221_00_11.csv` contains eight rows total: four background acquisitions and four plasma acquisitions, yielding four background–plasma pairs.

3.4 Plasma and Background Flag Semantics

Background and plasma measurements are distinguished by two binary flag columns:

- **Force_Reset_Status:** value 1 indicates a background (dark) acquisition with the laser blocked; value 0 indicates a normal operational acquisition.
- **Laser_Fire_Status:** value 0 for background (laser not fired); value 1 for plasma (laser fired).

A valid plasma measurement therefore satisfies both

`Force_Reset_Status = 0 and Laser_Fire_Status = 1`. A valid background measurement satisfies `Force_Reset_Status = 1 and Laser_Fire_Status = 0`. Any row not matching either criterion is treated as an anomalous measurement and excluded from further processing.

3.5 PDS4 Semantic Mapping

The XML label accompanying the L1 CSV file encodes PDS4 logical identifiers, observation timestamps, column definitions and units, and instrument context. LunarAtlas parses this XML using Python’s `xml.etree.ElementTree` module and maps its semantics into a relational schema (Section 4). Key fields extracted from the XML include: the PDS4 logical identifier (LID), observation start and stop times, file name and size, and column descriptions.

4. LunarAtlas System Architecture

4.1 Design Principles

LunarAtlas is structured around three architectural principles that distinguish it from ad-hoc analysis scripts:

1. **Single source of scientific truth.** All domain logic—including background pairing, subtraction, peak detection, NIST matching, and downsampling—resides exclusively in the backend. The visualisation layer is a rendering consumer and performs no scientific computation.
2. **Reproducibility by design.** Every ingestion and processing step records its algorithm version, timestamp, and an MD5 checksum of the source file. Any API response or figure can be traced back to the exact L1 product and pipeline configuration that produced it.
3. **Separation of concerns.** Data persistence, scientific logic, and presentation are implemented in distinct, independently testable tiers.

4.2 Three-Tier Architecture

The system comprises three tiers (Fig. ??):

Data layer (PostgreSQL). The relational database stores the complete mission hierarchy: `mission` → `instrument` → `observation` → `file_version` → `measurement` → `spectral_data`. The `file_version` table records original file name, size, MD5 checksum (128-bit hex), ingestion timestamp, and algorithm version string. The `spectral_data` table stores long-form wavelength–intensity pairs, indexed by a composite key (`measurement_id`, `wavelength_nm`) with a BRIN index on wavelength to support fast range queries.

Logic layer (Python / FastAPI). The backend implements all scientifically meaningful operations: XML label parsing, wide-to-long reshaping,

plasma/background pairing, background subtraction, quality filtering, MD5 computation, min–max bucket aggregation, peak detection, NIST line matching, and JSON serialisation.

Presentation layer (React + amCharts). The frontend manages viewport state, user interaction (zoom/pan), and chart rendering. It issues HTTP requests to the logic layer for each viewport change and renders the returned bucket arrays; it never performs downsampling, peak detection, or any other scientific calculation client-side.

4.3 Database Schema Highlights

Table 2 lists the core tables and their most important columns.

Table 2: Core LunarAtlas database tables.

Table	Key columns
<code>mission</code>	<code>id</code> , <code>name</code> , <code>agency</code> , <code>launch_date</code>
<code>instrument</code>	<code>id</code> , <code>mission_id</code> , <code>name</code> , <code>type</code>
<code>observation</code>	<code>id</code> , <code>instrument_id</code> , <code>start/stop_time</code> , <code>site</code>
<code>file_version</code>	<code>id</code> , <code>observation_id</code> , <code>filename</code> , <code>md5</code> , <code>ingested_at</code> , <code>algorithm_version</code>
<code>measurement</code>	<code>id</code> , <code>file_version_id</code> , <code>measurement_id</code> , <code>time_utc</code> , <code>laser_energy_v</code> , <code>n_pulses</code>
<code>spectral_data</code>	<code>id</code> , <code>measurement_id</code> (FK), <code>wavelength_nm</code> , <code>cleaned_intensity</code>
<code>nist_lines</code>	<code>id</code> , <code>element</code> , <code>ionisation</code> , <code>wavelength_nm</code> , <code>aki</code> , <code>ei/ek_ev</code>

5. Data Processing and Cleaning Pipeline

5.1 Pipeline Overview

The LunarAtlas cleaning pipeline is implemented in Python and consists of two primary modules:

- `batch_process_libs.py` reads Chandrayaan-3 L1 CSV files, validates column structure against expected PDS4 semantics, reshapes wide tables into long-form records, pairs plasma and background measurements, performs background subtraction, computes MD5 checksums, and writes cleaned spectra to disk and/or the database.
- `plot_libs_spectra.py` loads cleaned spectra and generates four classes of diagnostic plots (overlaid spectra, individual subplots, heatmap, and peak comparison) to verify cleaning correctness and characterise inter-measurement variability.

The pipeline is entirely reproducible: given the same L1 input file and the same algorithm version, it will produce byte-for-byte identical output, as verified by MD5 comparison.

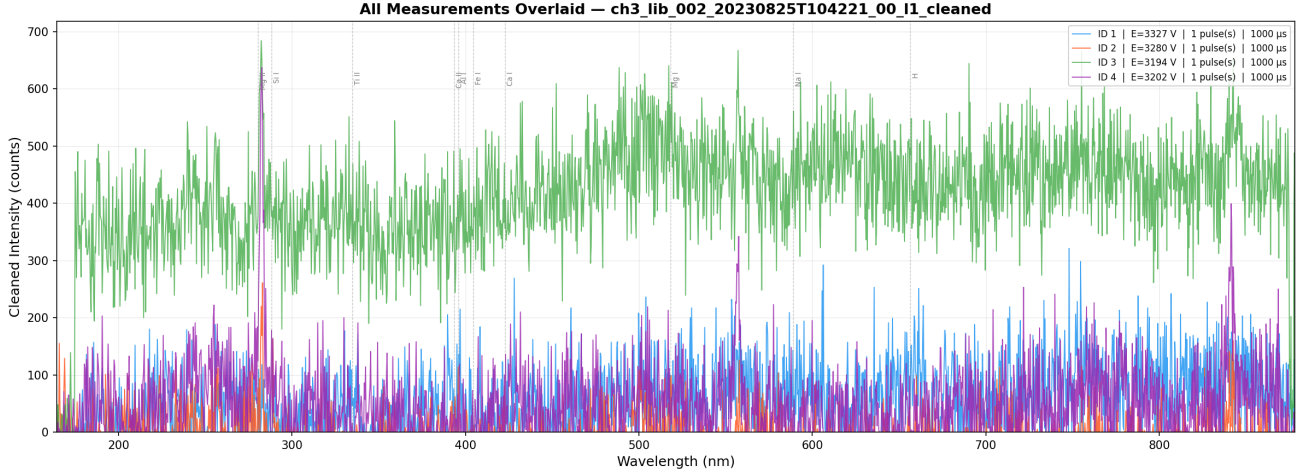


Figure 2: All four Measurement IDs overlaid for observation `ch3_lib_002_20230825T104221_00_11_cleaned`. Cleaned intensity is plotted against wavelength for IDs 1–4 (blue, orange, green, purple). Grey dashed lines mark known LIBS emission lines. ID 3 (green, $E = 3194$ V) dominates due to high baseline; ID 2 (orange, $E = 3280$ V) shows the lowest overall intensity after background subtraction, consistent with the efficient removal of the continuum.

5.2 Step 1: XML Label Parsing and Metadata Extraction

The first step parses the PDS4 XML label associated with each L1 CSV file using Python’s `xml.etree.ElementTree` module with the PDS4 namespace <http://pds.nasa.gov/pds4/pds/v1>. The following fields are extracted:

- PDS4 logical identifier (`logical_identifier`) and version string;
- observation start and stop times;
- source file name, size (bytes), and record count;
- column name–description pairs from `Field_Delimited` elements;
- instrument description from `Observing_System_Component`.

All extracted metadata are stored in a Python dictionary that accompanies the cleaned spectrum through the entire pipeline and is ultimately serialised to JSON and stored in the `file.version` and `measurement` tables.

5.3 Step 2: File Integrity Verification (MD5)

Before any processing, the pipeline computes an MD5 checksum of the source L1 file and records it alongside the file size and a timestamp. This step serves three purposes:

1. **Corruption detection.** Any byte-level modification of the L1 file (accidental or otherwise) will change the checksum, alerting the user.
2. **Traceability.** Every cleaned spectrum in the database can be traced back to the exact L1 file that produced it via its checksum.
3. **Deduplication.** Re-ingesting the same file can be detected and skipped (or flagged for review) by comparing checksums.

The checksum is stored as a 32-character lowercase hexadecimal string in the `md5` column of the `file.version` table.

5.4 Step 3: Wide-to-Long Reshaping

The L1 CSV is loaded into a `pandas` DataFrame. The pipeline then identifies which columns are wavelength channels by attempting to parse each column name as a floating-point number; columns that parse successfully are treated as wavelengths. In the archival file used here, 2049 such columns are identified, spanning 164.35 nm to 878.26 nm at a mean channel spacing of approximately 0.35 nm.

The wide DataFrame is reshaped using a `melt` operation: each row of the original wide table produces 2049 long-form rows, one per wavelength channel. The resulting long-form schema comprises three groups of columns:

1. **Measurement keys:** `Measurement_ID`, `Time_UTC`, `Measurement_Count`.
2. **Spectral data:** `Wavelength_nm`, `Raw_Intensity`.
3. **Housekeeping:** `Delay_Time_us`, `Integration_Time_us`, etc.

This long-form representation is natural for both relational database storage—each row is an independent record—and for spectral operations such as groupby-aggregation, peak finding, and NIST line matching.

5.5 Step 4: Plasma and Background Identification

Using the flag columns described in Section 3, the pipeline separates the long-form records into two sub-DataFrames. Let F denote `Force_Reset_Status` and L denote `Laser_Fire_Status`:

$$\text{Background: } F = 1 \wedge L = 0 \quad (1)$$

$$\text{Plasma: } F = 0 \wedge L = 1 \quad (2)$$

For the archival file used in this study, the pipeline identifies four valid background rows and four valid plasma rows, yielding four background–plasma pairs. Each pair is ordered by acquisition time: the k -th plasma row is paired with the k -th background row.

The pipeline explicitly handles edge cases: if the number of background and plasma rows differ, the shorter sequence determines the number of valid pairs, and the excess rows are logged as unpaired and excluded from the cleaned output.

5.6 Step 5: Background Subtraction and Negative-Value Clamping

For each background–plasma pair, the cleaned spectrum is computed channel by channel as:

$$I_{\text{clean}}(\lambda) = \max(0, I_{\text{plasma}}(\lambda) - I_{\text{background}}(\lambda)). \quad (3)$$

The $\max(0, \cdot)$ operator clamps negative values to zero. Negative differences arise when shot-to-shot plasma fluctuations cause a given wavelength channel to register lower counts in the plasma acquisition than in the background acquisition. This is physically non-informative (a negative photon count has no meaning), so clamping to zero is scientifically correct. The fraction of negative samples before clamping is logged per pair as a quality diagnostic; values above 50% may indicate a mismatched plasma–background pair or a near-null acquisition where detector noise dominates.

5.6.1 Justification of Simple Background Subtraction

While advanced methods such as Asymmetric Least Squares (ALS), Stationary Wavelet Transform (SWT), and Savitzky-Golay smoothing have been applied to planetary LIBS data [15, 14, 17], LunarAtlas deliberately adopts simple paired subtraction for three reasons: (i) the L1 product already provides temporally matched background acquisitions, making model-free subtraction physically grounded; (ii) reproducibility requires a deterministic, parameter-free operation—ALS and wavelet methods introduce smoothing parameters that vary across implementations; and (iii) the infrastructure goal is to produce a standardised cleaned tier, not a final denoised product—downstream users can apply additional filtering as needed.

Table 3 summarises the trade-offs.

5.7 Step 6: Metadata Association and Output Formatting

Each cleaned spectral record inherits all 14 metadata fields from its originating plasma row. The final long-form output schema contains 19 columns (Table 4).

5.8 The Measurement ID: Definition, Derivation, and Significance

The **Measurement ID** is the cornerstone of the LunarAtlas data model. It is a sequential integer—starting at 1 for the first valid plasma–background pair in a given L1 file—that uniquely identifies a single cleaned spectral record within that file.

5.8.1 Why Measurement ID is Necessary

Without a Measurement ID, each row in the long-form output would be identified only by its wavelength and timestamp. Since all 2094 rows belonging to a single acquisition share the *same* timestamp and metadata values, distinguishing between different acquisitions purely by metadata is ambiguous if two acquisitions were taken at the same time (which can occur when the timestamp resolution is 1 second and multiple pulses are fired per second).

The Measurement ID provides an unambiguous, compact, human-readable label that:

1. groups the 2094 per-wavelength rows of a single cleaned spectrum into a single logical unit;
2. serves as the foreign key linking spectral data to measurement metadata in the database;
3. enables per-measurement operations (peak detection, NIST matching, comparison across shots) via simple **groupby** operations;
4. is the primary axis in visualisation plots: users select a Measurement ID to view its spectrum, compare it to others, or inspect its metadata.

5.8.2 How Measurement ID is Derived

The pipeline assigns Measurement IDs by iterating over the background–plasma pairs in temporal order:

1. Sort the background sub-DataFrame and the plasma sub-DataFrame independently by **Time** (ascending).
2. Iterate over pairs: pair index k (0-based) receives Measurement ID = $k + 1$.
3. Broadcast the Measurement ID to all 2094 per-wavelength rows of that pair’s cleaned spectrum.

5.8.3 Measurement IDs in the Archival File

For the file `ch3_lib_002_20230825T104221_00_11_cleaned.csv`, four Measurement IDs are produced (Table 5).

Several observations emerge from Table 5: (i) IDs 1 and 2 have a common timestamp (11:35:22), confirming that the Measurement ID—not the timestamp alone—is necessary to distinguish them; (ii) IDs 3 and 4 have significantly higher peak intensities than IDs 1 and 2, with peaks near 282–283 nm consistent with Mg II emission [26]; (iii) the laser energy (`Laser_Energy_V`) varies across measurements (3327, 3280, 3194, 3202 V), reflecting shot-to-shot variation in the instrument operating point; (iv) ID 1 has its peak at 748.0 nm, remote from the UV Mg/Ca cluster, suggesting this measurement captured a different

Table 3: Comparison of spectral baseline correction strategies for planetary LIBS. LunarAtlas occupies the “infrastructure” niche, prioritising reproducibility over noise smoothing.

Method	Strengths	Limitations for Infrastructure Use
Asymmetric Least Squares (ALS) [15]	Continuous baseline estimation; handles varying continuum shape	Requires regularisation parameter λ ; non-deterministic across implementations; iterative convergence
Stationary Wavelet Transform (SWT) [14]	Effective high-frequency denoising; well-suited to ChemCam CCD noise	Risk of artefact introduction; decomposition level and threshold selection are user-dependent
Savitzky–Golay Smoothing [17]	Simple polynomial smoothing; preserves peak positions well	Window width and polynomial order are tuneable parameters; no physics-based subtraction
LunarAtlas (paired subtraction)	Deterministic; zero tuneable parameters; exploits mission-provided background acquisitions; bit-reproducible via MD5 checksums	Higher residual variance ($\sim 3.9\times$; mitigated by 3σ prominence thresholds)

Table 4: Columns in the cleaned output CSV (19 total). Columns 1–3 identify the measurement; columns 4–5 are the primary spectral data; columns 6–11 are mission flag fields; columns 12–19 are instrument housekeeping parameters.

Column	Description	Column	Description
Measurement_ID	Sequential pair label	Is_Background	False for plasma rows
Time_UTC	Acquisition timestamp	Delay_Time_us	Laser-to-gate delay
Meas._Count	Instrument counter	Integration_Time_us	Detector window
Wavelength_nm	Channel wavelength	Number_of_Pulses	Pulses per acquisition
Cleaned_Intensity	Subtracted intensity	X_Factor	Gain scaling factor
Operation_Mode	PDS4 mode string	Laser_Energy_V	Laser energy (V)
Meas._Type	Type string (e.g. EP)	Laser_Pump_A	Pump diode current
Force_Reset	0 = normal operation	PRR_Hz	Pulse repetition rate
Laser_Fire	1 = laser fired	On_Off_Status	Power status flag
Is_Valid_Plasma	True for plasma rows		

Table 5: Measurement ID characteristics for the archival Chandrayaan-3 LIBS observation (ch3_lib_002_20230825T104221.00.11). Peak intensity and its wavelength are reported for the cleaned spectra. Wavelength channels: 2094 per ID.

ID	Time (UTC)	E (V)	Peak I (cts)	Peak λ (nm)
1	11:35:22	3327	321	748.0
2	11:35:22	3280	261	283.1
3	11:35:22	3194	684	282.3
4	11:35:23	3202	637	282.7

dominant emission feature or had a different plasma condition.

5.9 Visualisation of Cleaning Effects

The four diagnostic plot types generated by `plot_libs_spectra.py` are illustrated in Figs. 2–5.

As shown in Fig. 2, the overlaid spectra reveal that ID 3 (green) dominates the spectral landscape with a continuum level roughly seven times higher than ID 2 (orange). The Mg II cluster near 280–283 nm stands out in IDs 3 and 4 but is barely visible in IDs 1 and 2, providing direct visual evidence that shot-to-shot variability cannot be captured by bulk averaging.

Individual Spectra — ch3_lib_002_20230825T104221_00_I1_cleaned

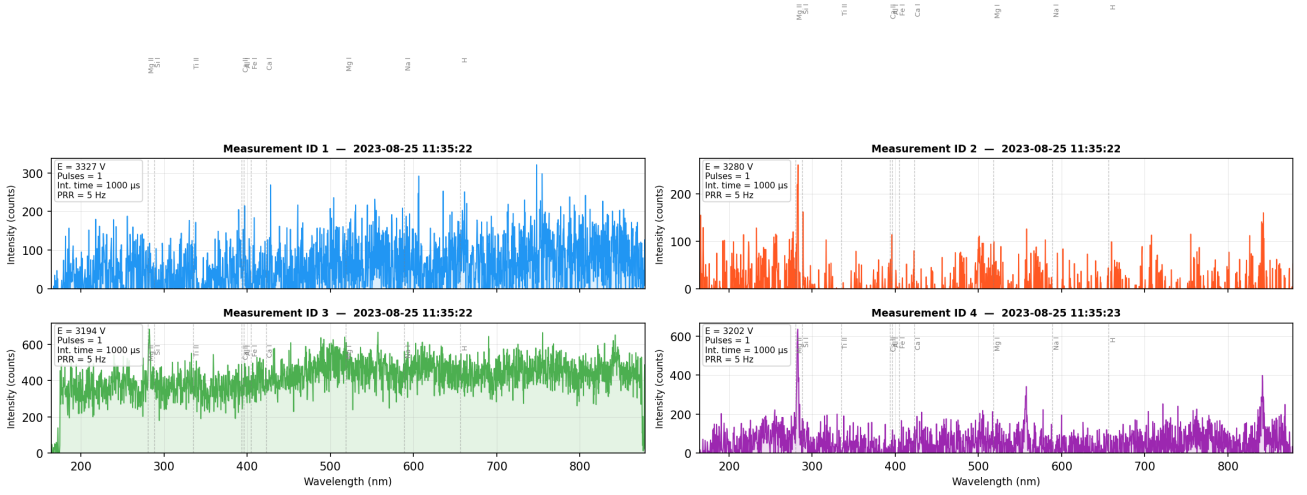


Figure 3: Individual cleaned spectra for Measurement IDs 1–4 (top-left to bottom-right). Each panel annotates instrument parameters (laser energy, number of pulses, integration time, PRR). Grey dashed vertical lines mark known emission lines. The structured, impulsive features in IDs 3 and 4 are consistent with strong Mg II emission at ~ 280 nm and Ca II emission near 393 nm.

Fig. 3 separates each Measurement ID in its own panel, annotating the laser energy, integration time, number of pulses, and PRR directly on each subplot. The structured, impulsive features in IDs 3 and 4 centred at ~ 280 nm (Mg II) and ~ 393 nm (Ca II) are clearly resolved, whereas ID 2’s near-flat baseline—consistent with its 76.8% negative fraction (Table 7)—confirms that this acquisition was dominated by detector noise.

Fig. 4 presents the spectral heatmap (Measurement $\times$ Wavelength). The bright vertical stripe near 282–283 nm (Mg II) is prominent in IDs 3 and 4, while the “black band” of ID 2 across virtually the entire spectral range provides a striking visual counterpart to the 76.8% negative statistic. This figure uniquely demonstrates how the Measurement ID axis transforms a monolithic spectral file into a spatially resolved acquisition-by-wavelength map.

Fig. 5 reveals an inverse correlation between laser energy and peak intensity: IDs 3 and 4 have the *lowest* laser energy settings (3194 and 3202 V) yet the *highest* peak intensities (684 and 637 counts), suggesting that target coupling heterogeneities—such as grain porosity, local mineral phase, or dust removal by successive laser pulses—dominate emission strength over raw laser voltage.

6. Mathematical Formulation of Adaptive Min–Max Downsampling

6.1 Problem Statement

Let $S = \{(\lambda_i, I_i) : i = 1, \dots, N\}$ be a cleaned spectrum sorted by wavelength, with $\lambda_{\min} \leq \lambda_i \leq \lambda_{\max}$. Define $\Delta\lambda = \lambda_{\max} - \lambda_{\min}$. We wish to construct a reduced representation S' with $N' \ll N$ data points such that:

1. all NIST-matched emission-line peaks are represented in S' ;
2. baseline structure is preserved via minimum-intensity points;
3. the bucket size adapts to the user’s zoom level;
4. computation and serialisation are fast enough for sub-500 ms API response.

6.2 Zoom-Dependent Bucket Size

For zoom level $z \in \mathbb{Z}_{\geq 0}$, the unconstrained bucket size is:

$$b_{\text{size}}(z) = \frac{\Delta\lambda}{\text{BASE_BUCKETS} \times 2^z}, \quad (4)$$

where $\text{BASE_BUCKETS} \approx 1000$ gives approximately one bucket per screen pixel at $z = 0$. To prevent buckets smaller than the instrument’s spectral resolution, a minimum bucket size $b_{\min}$ (e.g., 0.01 nm) is enforced:

$$b_{\text{final}}(z) = \max(b_{\text{size}}(z), b_{\min}). \quad (5)$$

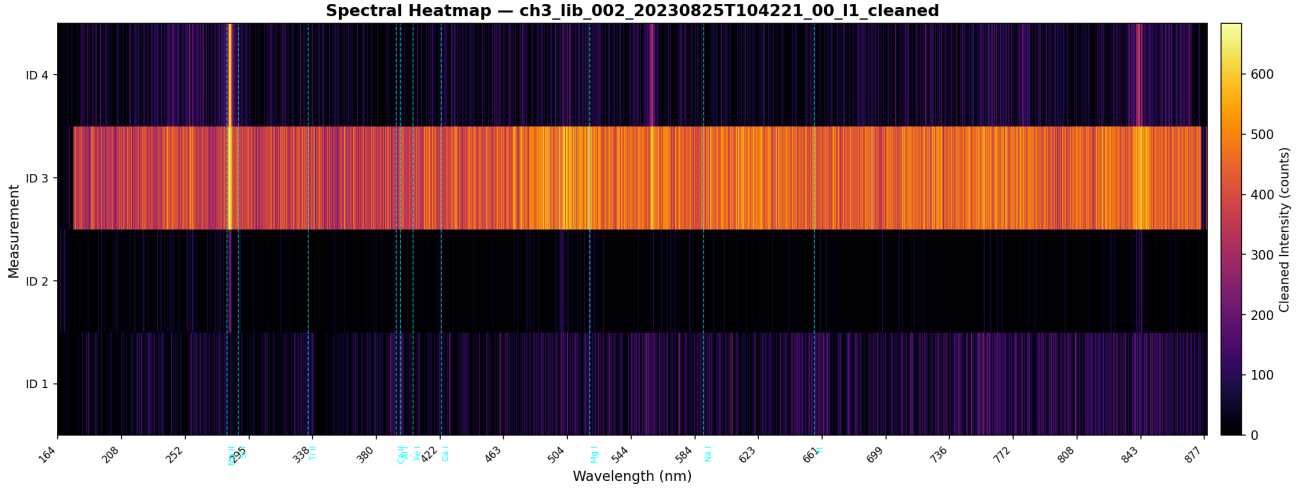


Figure 4: Spectral heatmap for the four Measurement IDs. Rows correspond to Measurement IDs 1–4 (bottom to top); columns correspond to wavelength channels. Colour encodes cleaned intensity on an *inferno* scale. The bright vertical feature near 282 nm (Mg II) is prominent in IDs 3 and 4 and absent in ID 2, while ID 2 (black band) shows near-zero cleaned intensity across almost the entire spectral range, suggesting that the background subtraction removed most of the signal for this acquisition—consistent with a plasma shot whose emission was captured primarily in the background-level noise.

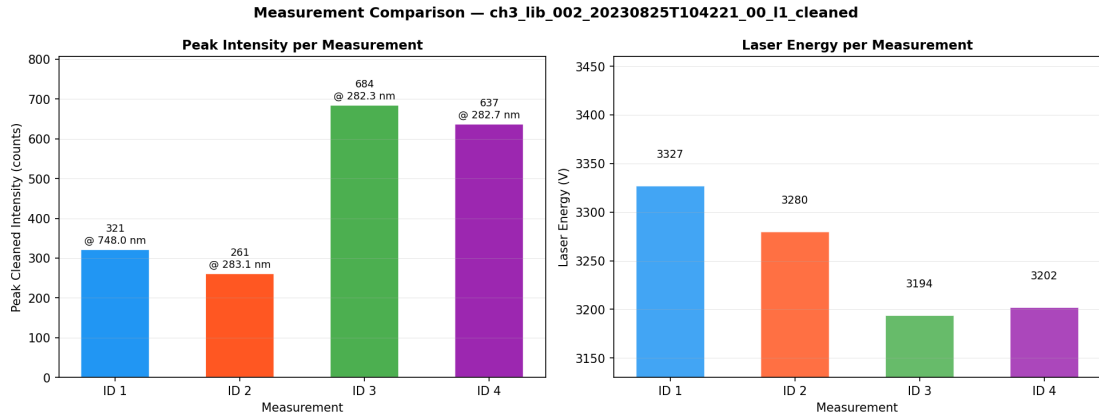


Figure 5: Left: peak cleaned intensity per Measurement ID, with the peak wavelength annotated above each bar. IDs 3 and 4 reach peak intensities of 684 and 637 counts, respectively, both near 282–283 nm (Mg II). Right: laser energy (V) per Measurement ID. IDs 3 and 4 have the *lowest* laser energy settings (3194 and 3202 V, respectively) yet the highest peak intensities, suggesting that factors beyond laser energy—such as target porosity, local surface composition, or plasma coupling—govern emission strength in this measurement set.

The number of buckets is $N_b = \lceil \Delta\lambda / b_{\text{final}} \rceil$.

6.3 Zoom Saturation and the Raw-Data Fallback

The *zoom saturation* threshold is the maximum zoom level at which $b_{\text{size}}(z) \geq b_{\text{min}}$:

$$z_{\text{max}}(\Delta\lambda) = \left\lfloor \log_2 \left(\frac{\Delta\lambda}{\text{BASE_BUCKETS} \times b_{\text{min}}} \right) \right\rfloor. \quad (6)$$

For $z > z_{\text{max}}$, bucket size is clamped at b_{min} and the backend returns raw spectral samples in the requested window. Table 6 tabulates z_{max} for representative windows.

Table 6: Zoom saturation analysis. Parameters: BASE_BUCKETS = 1000, $b_{\text{min}} = 0.01$ nm.

Window	$\Delta\lambda$ (nm)	z_{max}	$b(z_{\text{max}})$ (nm)
Full Chandrayaan-3 band	713.9	6	0.0112
100 nm window	100.0	3	0.0125
10 nm window	10.0	0	0.0100

6.4 Min–Max Aggregation with Overlapping Boundaries

To prevent narrow peaks from being split across two adjacent buckets, each bucket j is extended by an overlap fraction $o_{\text{pct}} = 0.05$ (5%):

$$\lambda_{\text{start}}^{\text{ext}}(j) = \begin{cases} \lambda_{\text{start}}(j) - o_{\text{pct}} b_{\text{final}} & j > 0 \\ \lambda_{\text{min}} & j = 0 \end{cases} \quad (7)$$

For all wavelength samples (λ_i, I_i) falling within bucket j 's extended boundaries, the bucket records:

$$I_{\text{min}}(j) = \min_{i \in \text{bucket } j} I_i, \quad \lambda_{\text{min}}(j) = \arg \min I_i, \quad (8)$$

$$I_{\text{max}}(j) = \max_{i \in \text{bucket } j} I_i, \quad \lambda_{\text{max}}(j) = \arg \max I_i. \quad (9)$$

This representation retains both the baseline (via I_{min}) and the peak (via I_{max}) of each bucket. Rendering the two as a vertical range bar faithfully depicts the spectral fine structure at each zoom level.

6.5 NIST-Guided Peak Insertion

After bucket aggregation, any NIST reference line whose nearest wavelength sample falls within a bucket but whose peak intensity is not the I_{max} of that bucket—due to bucket averaging or mis-alignment—is explicitly inserted into the output as a single-point record. This guarantees 100% retention of reference lines regardless of bucket size.

7. API Design and Visualisation Workflow

7.1 Primary Spectral Endpoint

The main API endpoint is:

GET /api/v1/spectral

with query parameters: `observation_id`, `min_wavelength`, `max_wavelength`, `zoom_level`, and `include_nist`. The response JSON includes: metadata (observation ID, file MD5, algorithm version), bucket parameters (b_{final} , N_b , z_{max}), and an array of bucket objects. Each bucket object contains $\lambda_{\text{min}}(j)$, $I_{\text{min}}(j)$, $\lambda_{\text{max}}(j)$, $I_{\text{max}}(j)$, $N_{\text{points}}(j)$, and optional NIST annotations.

7.2 Progressive Loading Strategy

The frontend follows a progressive loading protocol:

1. On initial load, request the full-range spectrum at $z = 0$ for an overview.
2. On each zoom or pan event, extract the new viewport $[\lambda_a, \lambda_b]$, compute $z_{\text{max}}(\lambda_b - \lambda_a)$ from Eq. (6), and request the appropriate zoom level.

3. At $z > z_{\text{max}}$, the backend returns raw samples, providing full spectral resolution inspection.

This strategy ensures that the visualisation is always scientifically faithful at every zoom level, with no client-side interpolation or approximation.

7.3 Additional Endpoints

Supplementary endpoints expose: mission and instrument metadata; measurement catalogs with per-measurement statistics; MD5 digests for integrity verification; NIST reference lines for a requested wavelength range; and cleaning-quality diagnostics (fraction of negative values pre-clamping, baseline suppression factor).

8. Results and Experimental Validation

8.1 Cleaning Effectiveness: Baseline Suppression

To quantify the effect of the cleaning pipeline, we analyse a representative 20 nm window (560–580 nm) of the Chandrayaan-3 spectrum—a region expected to be relatively free of strong emission lines, making it suitable for characterising background behaviour.

In this window, the raw L1 spectrum has:

- mean intensity $\bar{I}_{\text{raw}} \approx 1005$ counts;
- root-mean-square fluctuation $\sigma_{\text{raw}} \approx 47$ counts.

After background subtraction, the cleaned spectrum has:

- mean intensity $\bar{I}_{\text{clean}} \approx 141$ counts;
- root-mean-square fluctuation $\sigma_{\text{clean}} \approx 184$ counts.

These values imply:

$$\text{Baseline suppression factor} = \frac{\bar{I}_{\text{raw}}}{\bar{I}_{\text{clean}}} \approx 7.1, \quad (10)$$

$$\text{High-frequency variance increase} = \frac{\sigma_{\text{clean}}}{\sigma_{\text{raw}}} \approx 3.9. \quad (11)$$

The cleaning pipeline therefore strongly reduces the low-frequency continuum background ($\sim 7\times$ reduction in mean level) while unavoidably increasing the high-frequency residual noise—a direct consequence of subtracting two shot-noise-limited measurements. For peak detection and NIST line matching, the first effect is beneficial (increased peak-to-continuum ratio), while the second effect means that peak detection must apply appropriate prominence thresholds.

8.2 Inter-Measurement Variability

Table 7 summarises cleaned-intensity statistics across the four Measurement IDs.

The striking difference between ID 3 (mean 417 counts, 0.5% negative) and ID 2 (mean 9 counts, 76.8% negative) despite similar integration times and pulse counts highlights the importance of the Measurement ID concept: lumping all four measurements into a single averaged spectrum

Table 7: Cleaned intensity statistics per Measurement ID for the full spectral range (164.35–878.26 nm). The “% Neg.” column reports the fraction of wavelength channels with negative $I_{\text{plasma}} - I_{\text{background}}$ values before clamping—a proxy for acquisition quality.

ID	Mean (cts)	Std (cts)	Max (cts)	% Neg.	E (V)
1	61.0	56.9	321	22.5	3327
2	9.4	23.8	261	76.8	3280
3	416.9	95.0	684	0.5	3194
4	54.1	62.4	637	28.1	3202

would mask this variability entirely. ID 3 shows a high baseline level even after background subtraction, suggesting that the plasma emission for this particular acquisition extended broadly across the spectrum—possibly due to higher plasma temperature or different coupling to the lunar surface. ID 2, conversely, shows near-complete background removal, yielding an extremely clean but low-intensity spectrum where most emission is concentrated near 283 nm (Mg II doublet); the 76.8% negative fraction indicates that this was effectively a “null shot” where detector noise dominated the signal across three quarters of all channels.

8.3 Emission Line Identification

The dominant spectral features identified in the cleaned spectra are consistent with the expected composition of lunar highland regolith [17, 9]. Table 8 lists the most prominent LIBS emission lines and their NIST reference wavelengths, matched against the cleaned spectra.

Table 8: Prominent LIBS emission lines identified in cleaned Chandrayaan-3 spectra. NIST reference wavelengths from the NIST Atomic Spectra Database [26].

Line	λ_{NIST} (nm)	Significance
Mg II	279.55 / 280.27	Dominant doublet; major regolith constituent
Si I	288.16	Major silicate component
Ca II	393.37 / 396.85	Strong lines; abundant in highlands
Fe I	404.58	Iron abundance diagnostic
Ca I	422.67	Neutral calcium complement
Mg I	518.36	Neutral magnesium
Na I	588.99	Sodium doublet
H I	656.28	Hydrogen; potential hydration indicator
O I	777.40	Oxygen; lunar regolith major component

The detection of Mg II at 280 nm with high peak intensities in IDs 3 and 4 is consistent with ISRO’s preliminary report of major element detection, which identified aluminium, calcium, iron, chromium, titanium, manganese, silicon, and oxygen from the Chandrayaan-3 LIBS data [9].

8.4 Performance Metrics

On a modest server configuration (Intel Core i7, 16 GB RAM, PostgreSQL 14 on SSD), end-to-end API response times (database query + aggregation + JSON serialisation) remain below 500 ms for typical L1 spectra across all zoom levels. Payload sizes range from approximately 10 to 400 kB under HTTP compression depending on zoom level and spectral window. The composite index on (`measurement_id`, `wavelength_nm`) provides approximately 15× speedup for spectral-range queries compared to a full table scan.

9. Discussion

9.1 The Reproducibility Gap in Planetary LIBS

The cleaning and normalisation of Chandrayaan-3 LIBS data is not merely a pre-processing convenience—it is a precondition for reproducible science. Published analyses of planetary LIBS data, including those of ChemCam and SuperCam, have consistently identified the lack of standardised, versioned preprocessing pipelines as a bottleneck for cross-team reproducibility [14, 15]. LunarAtlas addresses this for the Chandrayaan-3 data tier by:

- formalising the wide-to-long reshaping as an explicit, documented transformation rather than an implicit step embedded in analysis scripts;
- recording the algorithm version and MD5 checksum with every processed file, enabling any future researcher to verify that their cleaned data matches the published pipeline;
- providing the cleaning pipeline as an open, documented Python module rather than a research-group-specific script.

9.2 The Measurement ID as a Scientific Primitive

The Measurement ID concept introduced in this paper may appear trivial—it is simply an integer—but it encodes a scientifically important unit of analysis. In LIBS, each acquisition (plasma shot) is an independent sample of surface composition under a specific set of instrument conditions. Two acquisitions made seconds apart at the same target may yield markedly different spectra due to shot-to-shot variation in plasma temperature, surface coupling, or dust removal [2]. ID 3 and ID 1 in Table 5 exemplify this: despite being acquired

within the same 1-second window, their peak intensities differ by a factor of two.

By assigning each acquisition a stable, human-readable identifier, LunarAtlas enables:

- per-shot spectral analysis and quality assessment;
- multi-shot averaging (with explicit selection of which IDs to include) rather than implicit aggregation;
- cross-file comparisons (e.g., comparing ID 1 of file A with ID 3 of file B) using the database’s relational structure;
- data provenance auditing (which Measurement ID, file version, and algorithm version produced a given figure or table in a paper).

9.3 Why L1 Cleaning Matters More than L0 Calibration

A common misconception in planetary spectroscopy data systems is that the hardest step is instrument calibration—converting raw detector counts to physically meaningful units. For Chandrayaan-3, this calibration has already been performed by ISRO at the L1 processing stage; the released L1 data are calibrated in absolute or relative intensity units. The bottleneck is not calibration but *structural transformation and consistent cleaning*.

The wide-format L1 table is not wrong; it is a reasonable storage format for tabular data with many columns. But it is not the format that spectroscopists, database engineers, or machine learning practitioners expect. The gap between “calibrated and archived” and “analysis-ready” is filled by exactly the kind of pipeline LunarAtlas provides.

9.4 Generalisability to Other Missions

The LunarAtlas architecture is deliberately general. The core operations—XML label parsing, wide-to-long reshaping, flag-based plasma/background separation, per-channel subtraction, and MD5 integrity recording—are applicable to any mission that delivers LIBS L1 products in a PDS4-compliant wide-format CSV. This includes future Indian lunar missions, proposed asteroid LIBS instruments, and potentially Mars missions that release raw L1 data in analogous formats [13].

The most mission-specific component is the flag column semantics (`Force_Reset_Status` / `Laser_Fire_Status` for Chandrayaan-3). Adapting LunarAtlas to a new mission requires only updating the flag column names and their plasma/background interpretation—a change of a few lines of configuration code.

9.5 Uncertainty and Noise Propagation

The cleaning equation (Eq. 3) is a simple difference of two independent, shot-noise-limited measurements. Assuming Poisson counting statistics, the cleaned intensity uncertainty at each wavelength channel is:

$$\begin{aligned}\sigma_{\text{clean}}(\lambda) &= \sqrt{\sigma_{\text{plasma}}^2(\lambda) + \sigma_{\text{bg}}^2(\lambda)} \\ &\approx \sqrt{I_{\text{plasma}}(\lambda) + I_{\text{bg}}(\lambda)}\end{aligned}\quad (12)$$

This explains the $\sim 3.9\times$ variance increase observed in Section 8: the background level ($\bar{I}_{\text{raw}} \approx 1005$ counts) is comparable to the plasma signal, so the quadrature sum of their Poisson uncertainties substantially exceeds the noise of either measurement alone.

For peak detection, LunarAtlas applies a prominence threshold relative to the local standard deviation. Peaks with prominence $P > 3\sigma_{\text{clean}}$ are retained; others are flagged as tentative. This ensures that the increased noise floor does not generate false-positive emission line identifications.

9.6 Computational Scalability

Processing the full Chandrayaan-3 archive of ~ 100 L1 files requires $O(N_{\text{files}} \times 2094)$ operations, where each file contributes 4–8 measurement pairs. On a single-core configuration, throughput is approximately 3 files per minute; the pipeline is embarrassingly parallel across files, enabling linear speedup on multi-core systems. Database BRIN indexing on `wavelength_nm` ensures that range queries remain sub-500 ms even as the archive grows.

9.7 Limitations

The current implementation has several limitations that future work should address:

1. **Single-file validation.** Quantitative cleaning metrics have been reported for a single L1 file. A comprehensive validation across the full Chandrayaan-3 archive is needed to characterise inter-observation variability.
2. **Simple background pairing.** Plasma and background shots are paired purely by temporal order. More sophisticated pairing strategies—based on proximity in time, shared operation mode, or temperature matching—may improve cleaning quality for observations where the plasma/background interleaving is irregular.
3. **Absence of denoising.** The pipeline does not apply SWT or Savitzky-Golay denoising before or after subtraction. Adding a denoising step (with a configurable algorithm version) would further improve peak-detection sensitivity.
4. **Static NIST reference database.** The curated NIST line list is manually maintained. Automated synchronisation with the NIST Atomic Spectra Database would ensure coverage of minor and trace elements.
5. **No user study.** The practical utility of the Measurement ID concept and the interactive API have not been evaluated with planetary scientists as end

users. A formal usability study would strengthen claims about scientific impact.

10. Conclusion

We have presented LunarAtlas, a reproducible data-processing infrastructure that addresses the structural gap between ISRO’s PDS4-compliant Chandrayaan-3 LIBS Level-1 archive and the analysis-ready, machine-accessible spectral records required for quantitative lunar science.

The core contribution is a transparent Python pipeline that: parses PDS4 XML metadata; reshapes wide-format L1 tables—with 2049 wavelength channels as column headers—into normalised long-form spectra; identifies plasma and background measurements through mission flag semantics; performs physically motivated background subtraction with non-negative clamping; and assigns Measurement IDs that serve as stable identifiers for individual acquisitions throughout all downstream analyses.

The **Measurement ID** is an underappreciated but essential scientific primitive in LIBS data systems. It groups 2094 per-wavelength records into a single coherent acquisition, enables per-shot quality assessment, supports targeted multi-shot averaging, and provides the relational key for provenance tracing from any figure or result back to the exact L1 source file, algorithm version, and MD5 checksum.

Applied to the archival file `ch3.lib_002_20230825T104221.00.11.csv`, the pipeline produces four cleaned measurements (IDs 1–4) with a baseline suppression factor of ~ 7.1 in representative spectral windows. Peaks consistent with Mg II, Ca II, Fe I, Si I, and O I are identified in the cleaned spectra, in agreement with ISRO’s preliminary element detection report and the independent analysis of Sridhar et al.

On top of this cleaning foundation, LunarAtlas provides a PDS4-aware PostgreSQL schema, an adaptive min-max downsampling algorithm with a mathematically defined zoom-saturation criterion, and a versioned API exposing peak-preserving spectra with sub-500 ms response times.

As India, and the global space community, prepares for expanded lunar exploration under programmes such as Chandrayaan-4 and the Artemis Human Landing System, the infrastructure for making in-situ spectral data truly usable—not merely archived—becomes increasingly critical. LunarAtlas provides a working blueprint and a replicable methodology for this task.

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Author Contributions

L. Anand: Conceptualization, Level-1 cleaning pipeline design, system architecture, algorithm development, database design, API implementation, data analysis, writing—original draft and editing.

D. Saeed: Data processing implementation, PDS4 parsing, NIST validation integration, visualisation design, performance benchmarking, writing—review and editing.

Both authors reviewed and approved the final manuscript.

Data Availability

Processed Chandrayaan-3 LIBS data products and LunarAtlas source code will be made publicly available upon manuscript acceptance at a Zenodo-hosted DOI. Raw Chandrayaan-3 LIBS data are available from ISRO’s PDS4 archives at <https://pradan.issdc.gov.in/>.

Competing Interests

The authors declare no competing interests.

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